MAUAJAMA FIRDAUS

Curriculum Vitae

12/1A, Pementle Street
Kolkata
(+91) 9674841004
mauajama@iitism.ac.in
Github *Linkedin*

Experience

Department of Computer Science and Engineering, Indian Institute of Technology (Indian School Of Mines) Dhanbad, India

June,2024 – Present ***Assistant Professor***.
Natural Language Processing, Multimodality and Multilinguality, Explainable AI

Department of Computing Science, University of Alberta, Canada

March,2022 – June, 2024 ***PostDoctoral Research Associate***.
Natural Language processing, Text Generation

Advisor : **Dr. Lili Mou**, *Assistant Professor, Department of Computing Science*, University of Alberta

Accenture Global Solutions Ltd., Bangalore, India

August,2017 – January,2018 ***Research Intern***.
Natural Language Understanding

Advisor : **Dr. Harshawardhan Madhukar Wabgaonkar, and Dr. Shubhashis Sengupta**

Education

2016–2021 **PhD, Computer Science & Engineering**, *Indian Institute of Technology*, Patna.
Natural Language Processing, Dialogue Systems, Natural language Understanding, Natural Language Generation, Multimodality, Machine Learning and Deep Learning

2014–2016: **Master of Engineering, Computer Science and Engineering**, *National Institute of Technology*, Durgapur.

2010–2014: **Bachelor of Engineering, Computer Science and Engineering**, *Narula Institute of Technology*, Kolkata.

Publications & Tutorial

Patent

2020 Harshawardhan Madhukar Wabgaonkar, Shubhashis Sengupta, Tirupal Rao Ravilla, Sumitraj Ganapat Patil, Poulami Debnath, GM Sushravya, Roshni Ramesh Ramnani, Gurudatta Mishra, Moushumi Mahato, **Mauajama Firdaus**, et al. Spoken language understanding system and method using recurrent neural networks, June 9 2020. US Patent 10,679,613.

Tutorial

2023 **Priyanshu Priya, Mauajama Firdaus, Kshitij Mishra and Asif Ekbal**, Empathetic Conversational Artificial Intelligence Systems: Recent Advance and New Frontiers, In The 32nd International Joint Conference on Artificial Intelligence (IJCAI), 2023.

2023 **Priyanshu Priya, Mauajama Firdaus, Gopendra Vikram Singh and Asif Ekbal**, Navigating the Landscape of Empathetic Dialogue Systems: Current Innovations and Future Developments, In The 20th International Conference on Natural Language Processing (ICON), 2023.

2024 **Priyanshu Priya, Mauajama Firdaus, Gopendra Vikram Singh and Asif Ekbal**, Affective Computing for Social Good Applications: Current Advances, Gaps and Opportunities in Conversational Setting, In The 46th European Conference on Information Retrieval (ECIR), 2024.

2024 **Priyanshu Priya, Deeksha Varshney, Mauajama Firdaus and Asif Ekbal**, Knowledge-enhanced Response Generation in Dialogue Systems: Current Advancements and Emerging Horizons, In the Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024), 2024.

Journal Articles

2025 Gopendra Vikram Singh, Soumitra Ghosh, **Mauajama Firdaus**, Asif Ekbal, and Pushpak Bhattacharyya. Unmasking offensive content: a multimodal approach with emotional understanding. *Multimedia Tools and Applications*, pages 1–24. Springer, 2025.

2025 Priyanshu Priya, Armita Mani Tripathi, Deeksha Varshney, **Mauajama Firdaus**, and Asif Ekbal. Empower-kare: Deep prompt learning for knowledge-aware response generation in clinical counseling and legal support conversations. *IEEE Transactions on Artificial Intelligence*. IEEE, 2025.

2024 Gopendra Vikram Singh, **Mauajama Firdaus**, Dushyant Singh Chauhan, Asif Ekbal, and Pushpak Bhattacharyya. Zero-shot multitask intent and emotion prediction from multimodal data: A benchmark study. *Neurocomputing*, volume 569, page 127128. Elsevier, 2024.

2024 Gopendra Vikram Singh, Soumitra Ghosh, **Mauajama Firdaus**, Asif Ekbal, and Pushpak Bhattacharyya. Predicting multi-label emojis, emotions, and sentiments in code-mixed texts using an emojifying sentiments framework. *Scientific Reports*, volume 14, pages 1–24. Nature Publishing Group, 2024.

2024 Priyanshu Priya, **Mauajama Firdaus**, and Asif Ekbal. Xeropol: Emotion-aware contrastive learning for zero-shot cross-lingual politeness identification in dialogues. *IEEE Transactions on Computational Social Systems*. IEEE, 2024.

2024 Priyanshu Priya, **Mauajama Firdaus**, and Asif Ekbal. Two in one: A multi-task framework for politeness turn identification and phrase extraction in goal-oriented conversations. *Computer Speech & Language*, page 101661. Elsevier, 2024.

2024 Priyanshu Priya, **Mauajama Firdaus**, and Asif Ekbal. Computational politeness in natural language processing: A survey. *ACM Computing Surveys*. ACM New York, NY, 2024.

2024 Kshitij Mishra, **Mauajama Firdaus**, and Asif Ekbal. Please donate to save a life: Inducing politeness to handle resistance in persuasive dialogue agents. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*. IEEE, 2024.

2023 **Mauajama Firdaus**, Gopendra Vikram Singh, Asif Ekbal, and Pushpak Bhattacharyya. Affect-gcn: a multimodal graph convolutional network for multi-emotion with intensity recognition and sentiment analysis in dialogues. *Multimedia Tools and Applications*, pages 1–22. Springer, 2023.

2023 **Mauajama Firdaus**, Avinash Madasu, and Asif Ekbal. A unified framework for slot based response generation in a multimodal dialogue system. *Multimedia Tools and Applications*. Springer, 2023.

2023 **Mauajama Firdaus**, Asif Ekbal, and Erik Cambria. Multitask learning for multilingual intent detection and slot filling in dialogue systems. *Information Fusion*, volume 91, pages 299–315. Elsevier, 2023.

2023 Priyanshu Priya, **Mauajama Firdaus**, and Asif Ekbal. A multi-task learning framework for politeness and emotion detection in dialogues for mental health counseling and legal aid. *Expert Systems with Applications*, page 120025. Elsevier, 2023.

2023 Kshitij Mishra, **Mauajama Firdaus**, and Asif Ekbal. Genpads: Reinforcing politeness in an end-to-end dialogue system. *Plos One*, volume 18, page e0278323. Public Library of Science San Francisco, CA USA, 2023.

2022 **Mauajama Firdaus**, Naveen Thangavelu, Asif Ekbal, and Pushpak Bhattacharyya. I enjoy writing and playing, do you: A personalized and emotion grounded dialogue agent using generative adversarial network. *IEEE Transactions on Affective Computing*. IEEE, 2022.

2022 **Mauajama Firdaus**, Arunav Shandilya, Asif Ekbal, and Pushpak Bhattacharyya. Being polite: Modeling politeness variation in a personalized dialog agent. *IEEE Transactions on Computational Social Systems*. IEEE, 2022.

2022 Gopendra Vikram Singh, **Mauajama Firdaus**, Shambhavi, Shruti Mishra, and Asif Ekbal. Knowing what to say: Towards knowledge grounded code-mixed response generation for open-domain conversations. *Knowledge-Based Systems*, volume 249, page 108900. Elsevier, 2022.

2022 Gopendra Vikram Singh, **Mauajama Firdaus**, Asif Ekbal, and Pushpak Bhattacharyya. Unity in diversity: Multilabel emoji identification in tweets. *IEEE Transactions on Computational Social Systems*. IEEE, 2022.

2022 Gopendra Vikram Singh, **Mauajama Firdaus**, Asif Ekbal, and Pushpak Bhattacharyya. Emointtrans: A multimodal transformer for identifying emotions and intents in social conversations. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, volume 31, pages 290–300. IEEE, 2022.

2022 Kshitij Mishra, **Mauajama Firdaus**, and Asif Ekbal. Predicting politeness variations in goal-oriented conversations. *IEEE Transactions on Computational Social Systems*. IEEE, 2022.

2022 Kshitij Mishra, **Mauajama Firdaus**, and Asif Ekbal. Please be polite: Towards building a politeness adaptive dialogue system for goal-oriented conversations. *Neurocomputing*. Elsevier, 2022.

2021 **Mauajama Firdaus**, Nidhi Thakur, and Asif Ekbal. Aspect-aware response generation for multimodal dialogue system. *ACM Transactions on Intelligent Systems and Technology (TIST)*, volume 12, pages 1–33. ACM New York, NY, USA, 2021.

2020 **Mauajama Firdaus**, Arunav Pratap Shandeelya, and Asif Ekbal. More to diverse: Generating diversified responses in a task oriented multimodal dialog system. *PloS one*, volume 15, page e0241271. Public Library of Science San Francisco, CA USA, 2020.

2020 **Mauajama Firdaus**, Hitesh Golchha, Asif Ekbal, and Pushpak Bhattacharyya. A deep multi-task model for dialogue act classification, intent detection and slot filling. *Cognitive Computation*, pages 1–20. Springer, 2020.

2020 **Mauajama Firdaus**, Hardik Chauhan, Asif Ekbal, and Pushpak Bhattacharyya. Emosen: Generating sentiment and emotion controlled responses in a multimodal dialogue system. *IEEE Transactions on Affective Computing*. IEEE, 2020.

2019 **Mauajama Firdaus**, Ankit Kumar, Asif Ekbal, and Pushpak Bhattacharyya. A multi-task hierarchical approach for intent detection and slot filling. *Knowledge-Based Systems*, volume 183, page 104846. Elsevier, 2019.

In Conference Proceedings

2025 Priyanshu Priya, Rishikant Chigrupaatii, **Mauajama Firdaus**, and Asif Ekbal. Genteel-negotiator: Llm-enhanced mixture-of-expert-based reinforcement learning approach for polite negotiation dialogue. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 25010–25018, 2025.

2024 Gopendra Vikram Singh, **Mauajama Firdaus**, Dushyant Singh Chauhan, Asif Ekbal, and Pushpak Bhattacharyya. Well, now we know! unveiling sarcasm: Initiating and exploring multimodal conversations with reasoning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 18981–18989, 2024.

2024 Gopendra Singh, Sai Vemulapalli, **Mauajama Firdaus**, and Asif Ekbal. Deciphering cognitive distortions in patient-doctor mental health conversations: A multimodal llm-based detection and reasoning framework. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 22546–22570, 2024.

2024 Sajad Ramezani, **Mauajama Firdaus**, and Lili Mou. Claim-centric and sentiment guided graph attention network for rumour detection. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 3235–3241, 2024.

2024 Priyanshu Priya, Gopendra Singh, **Mauajama Firdaus**, Jyotsna Agrawal, and Asif Ekbal. On the way to gentle ai counselor: Politeness cause elicitation and intensity tagging in code-mixed hinglish conversations for social good. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 4678–4696, 2024.

2023 Zijun Wu, Zi Xuan Zhang, Atharva Naik, Zhijian Mei, **Mauajama Firdaus**, and Lili Mou. Weakly supervised explainable phrasal reasoning with neural fuzzy logic. In *Proceedings of the Eleventh International Conference on Learning Representations (ICLR)*, 2023.

2023 **Mauajama Firdaus**, Gopendra Singh, Asif Ekbal, and Pushpak Bhattacharyya. Multi-step prompting for few-shot emotion-grounded conversations. In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*, pages 3886–3891, 2023.

2023 **Mauajama Firdaus**, Priyanshu Priya, and Asif Ekbal. Mixing it up: Inducing empathy and politeness using multiple behaviour-aware generators for conversational systems. In *Findings of the Association for Computational Linguistics: IJCNLP-AACL 2023 (Findings)*, pages 336–347, 2023.

2023 Guoqing Luo, Yutong Han, Lili Mou, and **Mauajama Firdaus**. Prompt-based editing for text style transfer. In *Findings of the Association for Computational Linguistics: EMNLP 2023, Singapore, December 6-10, 2023*, pages 5740–5750. Association for Computational Linguistics, 2023.

2022 **Mauajama Firdaus**, Nidhi Thakur, and Asif Ekbal. Sentiment guided aspect conditioned dialogue generation in a multimodal system. In *European Conference on Information Retrieval*, pages 199–214. Springer, 2022.

2022 **Mauajama Firdaus**, Asif Ekbal, and Pushpak Bhattacharyya. Polise: Reinforcing politeness using user sentiment for customer care response generation. In *Proceedings of the 29th International Conference on Computational Linguistics (COLING)*, pages 6165–6175, 2022.

2022 Gopendra Vikram Singh, Priyanshu Priya, **Mauajama Firdaus**, Asif Ekbal, and Pushpak Bhattacharyya. Emoinhindi: A multi-label emotion and intensity annotated dataset in hindi for emotion recognition in dialogues. *LREC*, 2022.

2022 Gopendra Vikram Singh, Dushyant Singh Chauhan, **Mauajama Firdaus**, Asif Ekbal, and Pushpak Bhattacharyya. Are emoji, sentiment, and emotion friends? a multi-task learning for emoji, sentiment, and emotion analysis. In *Proceedings of the 36th Pacific Asia Conference on Language, Information and Computation*, pages 166–174, 2022.

2022 Azlaan Mustafa Samad, Kshitij Mishra, **Mauajama Firdaus**, and Asif Ekbal. Empathetic persuasion: Reinforcing empathy and persuasiveness in dialogue systems. In *Findings of the Association for Computational Linguistics: NAACL 2022*, pages 844–856, 2022.

2021 **Mauajama Firdaus**, Nidhi Thakur, and Asif Ekbal. Multi-aspect controlled response generation in a multimodal dialogue system using hierarchical transformer network. In *2021 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8. IEEE, 2021.

2021 **Mauajama Firdaus**, Arunav Pratap Shandilya, Sthita Pragyan Pujari, and Asif Ekbal. Attribute centered multimodal response generation in a dialogue system. In *2021 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8. IEEE, 2021.

2021 **Mauajama Firdaus**, Umang Jain, Asif Ekbal, and Pushpak Bhattacharyya. Seprg: Sentiment aware emotion controlled personalized response generation. In *Proceedings of the 14th International Conference on Natural Language Generation*, pages 353–363, 2021.

2021 **Mauajama Firdaus**, Hardik Chauhan, Asif Ekbal, and Pushpak Bhattacharyya. More the merrier: Towards multi-emotion and intensity controllable response generation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 12821–12829, 2021.

2020 **Mauajama Firdaus**, Naveen Thangavelu, Asif Ekba, and Pushpak Bhattacharyya. Persona aware response generation with emotions. In *2020 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8. IEEE, 2020.

2020 **Mauajama Firdaus**, Nidhi Thakur, and Asif Ekbal. Multidm-gcn: Aspect-guided response generation in multi-domain multi-modal dialogue system using graph convolution network. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: Findings*, pages 2318–2328, 2020.

2020 **Mauajama Firdaus**, Asif Ekbal, and Pushpak Bhattacharyya. Incorporating politeness across languages in customer care responses: Towards building a multi-lingual empathetic dialogue agent. In *Proceedings of The 12th Language Resources and Evaluation Conference*, pages 4172–4182, 2020.

2020 **Mauajama Firdaus**, Hardik Chauhan, Asif Ekbal, and Pushpak Bhattacharyya. Meisd: A multimodal multi-label emotion, intensity and sentiment dialogue dataset for emotion recognition and sentiment analysis in conversations. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 4441–4453, 2020.

2019 Tulika Saha, Saurabh Srivastava, **Mauajama Firdaus**, Sriparna Saha, Asif Ekbal, and Pushpak Bhattacharyya. Exploring machine learning and deep learning frameworks for task-oriented dialogue act classification. In *2019 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8. IEEE, 2019.

2019 Hitesh Golchha, **Mauajama Firdaus**, Asif Ekbal, and Pushpak Bhattacharyya. Courteously yours: Inducing courteous behavior in customer care responses using reinforced pointer generator network. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 851–860, 2019.

2019 Hardik Chauhan, **Mauajama Firdaus**, Asif Ekbal, and Pushpak Bhattacharyya. Ordinal and attribute aware response generation in a multimodal dialogue system. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 5437–5447, 2019.

2018 **Mauajama Firdaus**, Shobhit Bhatnagar, Asif Ekbal, and Pushpak Bhattacharyya. Intent detection for spoken language understanding using a deep ensemble model. In *Pacific Rim international conference on artificial intelligence*, pages 629–642. Springer, 2018.

2018 **Mauajama Firdaus**, Shobhit Bhatnagar, Asif Ekbal, and Pushpak Bhattacharyya. A deep learning based multi-task ensemble model for intent detection and slot filling in spoken language understanding. In *International Conference on Neural Information Processing*, pages 647–658. Springer, 2018.

Research Experience

Indian Institute of Technology, Patna

December,2020 – 2021 ***Modeling Politeness Variation in a Personalized Dialogue Agent***.

Developing a novel Polite Personalized Dialogue Generation (PoPe-DG) framework that employs a reinforced deliberation network for generating polite personalized dialogue responses in accordance with the user profile and consistent with the conversational history.

Advisor : **Dr. Asif Ekbal**, *Associate Professor, Computer Science & Engineering*, IIT Patna & **Prof. Pushpak Bhattacharyya**, *Professor, Computer Science & Engineering*, IIT Bombay

July,2019 – October,2020 ***Aspect-aware Response Generation for Multimodal Dialogue System***.
In our current work, we focus on the task of aspect controlled response generation in a multimodal task-oriented dialog system. As there was no readily available data for building such multimodal systems, we create a Multi-Domain Multi-Modal Dialog (MDMMD++) dataset. The dataset comprises of the conversations having both text and images belonging to the four different domains, such as hotels, restaurants, electronics and furniture.

Advisor : **Dr. Asif Ekbal**, *Associate Professor, Department of Computer Science & Engineering*, IIT Patna

February,2019 – September,2020 ***Emotion and Sentiment controlled Dialogue Systems***.
Proposed an approach for sentiment and emotion controlled dialogue generation. Creation of large-scale emotion and sentiment controlled dialogue dataset. In addition, we introduce the task of multi-emotion controllable response generation with the ability to express different emotions with varying levels of intensity in an open-domain dialogue system.

Advisor : **Dr. Asif Ekbal**, *Associate Professor, Computer Science & Engineering*, IIT Patna & **Prof. Pushpak Bhattacharyya**, *Professor, Computer Science & Engineering*, IIT Bombay

June,2018 – December,2018 ***Inducing courteous phrases in Customer care Responses***.
In this work, we propose an effective deep learning framework for inducing courteous behavior in customer care responses. Our technique is based on a reinforced pointer-generator model for the sequence to sequence task. The model is also conditioned on a hierarchically encoded and emotionally aware conversational context. We use real interactions on Twitter between customer care professionals and aggrieved customers to create a large conversational dataset.

Advisor : **Dr. Asif Ekbal**, *Associate Professor, Computer Science & Engineering*, IIT Patna & **Prof. Pushpak Bhattacharyya**, *Professor, Computer Science & Engineering*, IIT Bombay

January,2017 – May,2018 ***Multi-task framework for Natural Language Understanding***.
Proposed an end-to-end hierarchical multi-task framework that can jointly perform both intent detection (ID) and slot filling (SF) tasks for the datasets of varying domains. Besides, we also propose a multi-task network that in addition to ID and SF also performs Dialogue Act Classification (DAC).

Advisor : **Dr. Asif Ekbal**, *Associate Professor, Computer Science & Engineering*, IIT Patna & **Prof. Pushpak Bhattacharyya**, *Professor, Computer Science & Engineering*, IIT Bombay

National Institute of Technology, Durgapur

July,2015 – May,2016 ***Constraint specification in Multi Agent System***.
This work proposes formal analysis method for defining and specifying properties, resources, knowledge and related constraints of an agent in a Multi Agent System (MAS).

Advisor : **Dr. Rajiv Kumar Chatterjee**, *Assistant Professor, Department of Computer Science & Engineering*, NIT Durgapur

Fellowships & Awards

2019 Receipt of ***Microsoft Travel Grant*** to attend the ***17th Annual Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT), 2019*** in Minneapolis, United States.

2019 Receipt of ***Microsoft Travel Grant*** to attend the ***57th Annual Meeting of the Association for Computational Linguistics (ACL), 2019*** in Florence, Italy.

2019 Receipt of the ***ACM India-IARCS travel grant*** to attend ***57th Annual Meeting of the Association for Computational Linguistics(ACL), 2020*** in Florence, Italy.

Computer skills

Programming Languages: Python, PyTorch, Keras, Tensorflow, C, C++, Advanced JAVA

Web Technologies: HTML 5, PHP, JSP, Javascript

Database: SQL, MySQL, Apache, Neo4j

Teaching

Monsoon, 2024 : **CSO303: Principles of Artificial Intelligence**, IIT(ISM) Dhanbad.

Monsoon, 2024 : **CSD411: Introduction to Deep Learning**, IIT(ISM) Dhanbad.

Winter, 2025 : **NCSV101: Computer Programming**, IIT(ISM) Dhanbad.

Winter, 2025 : **NCSV102: Computer Programming Lab**, IIT(ISM) Dhanbad.

Current Professional Services

Associate Editor: **Neural Networks, Elsevier**

; Associate Editor: **Computers and Electrical Engineering, Elsevier**

;

Research Grants

Faculty Research Scheme: **Principal Investigator** for the project titled *"Hate Speech Detection Across Languages and Modalities: Enhancing LLMs with Reward-Based Learning and Explainable AI"* under the **Faculty Research Scheme**, [IIT(ISM) Dhanbad]; **Funding Amount:** 20,00,000/- ; **Duration:** 2 years ; **Project Summary:** This project focuses on leveraging large language models with explainable AI and reinforcement learning to detect hate speech across languages and modalities.

Specialized Research Grant: Awarded a **Specialized Research Grant** to establish an **Advanced Laboratory for Multilingual and Multimodal AI Research** at [IIT(ISM) Dhanbad]; **Funding Amount:** 30,00,000/- ; **Grant Scope:** This initiative supports the development of infrastructure and research capacity for AI-NLP related research endeavors.

Referees

Dr. Asif Ekbal
Associate Professor, Department of Computer Science & Engineering
Indian Institute of Technology Patna
✉ asif.ekbal@gmail.com

Prof. M Tanveer
Professor, Department of Mathematics
Indian Institute of Technology Indore
✉ mtanveer@iiti.ac.in

Dr. Shubhashis Sengupta
Managing Director and Head of Research Accenture Labs, India
✉ shubhashis.sengupta@accenture.com

Dr. Lili Mou
Assistant Professor, Department of Computing Science
University of Alberta, Canada
✉ lmou@ualberta.ca